

Roman Young

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EDUCATION

University of California, San Diego | 4.00 GPA

La Jolla, CA | Expected Grad: 2028

B.S. Bioinformatics, Computer Science Minor

Relevant Coursework: Organic Chemistry (I & II), Data Analysis for Biologists (R), Advanced Bioinformatics Lab, Discrete Math (I & II), Advanced Data Structures, Linear Algebra, Probability and Statistics in Bioinformatics,

EXPERIENCE

Salk Institute for Biological Studies | Research Intern

La Jolla, CA | March 2025 - Present

- Advanced research on immune ontogeny by **profiling IgA-coated maternal microbiota**; identified key microbial targets that influence neonatal gut colonization, directly contributing to projects on developmental origins of immune health.
- Executed **gel electrophoresis** and genotyping for **Cre/lox-based transgenic mice** to confirm APC knockout efficiency.
- Carefully dissected and prepared **100+** colon/intestinal organs for antibody staining to isolate IEL, LPL, and Myeloid cells.

La Jolla Institute of Immunology | Project Lead

La Jolla, CA | Sept. 2025 - Present

- Leading the development of an indel-aware peptide matching pipeline, transitioning from substitution-only logic to a complex InDel-alignment model to improve biological realism and curation of the **IEDB** (Immune Epitope Database).
- Implemented a high-speed **k-mer** indexing system using **Python**, **Polars**, and a **C Hamming** function, reducing the search time for **100+ queries** against the full human proteome by up to **40,000x** from hours to sub-second execution.
- Designed **30** large-scale validation jobs via **Slurm** clusters, processing millions of alignments to verify **100% sensitivity**.

Kaiser Permanente | Doctor Shadow

Fremont, CA | Sept. 2023 - May 2024

- Analyzed clinical decision-making processes by shadowing an Internal Medicine specialist across **200+** patient encounters
- Gained insight into clinical decision-making by observing the administration of preventative immunizations (Polio, Influenza, Shingles) and the diagnostic evaluation of post-operative recovery through MRI and radiographic imaging
- Engaged in medical discussions about daily operations, instrument usage, and Kaiser's medication prescription software.

Great Shape International Dental Project | Clinical Assistant

San Jose, CA | Sept. 2023 - Present

- Hand-selected for international humanitarian missions to deliver dental care to underserved populations in the Caribbean
- Executed clinical dental treatments for **220+** patients, supporting a range of clinical procedures including diagnostic **X-rays** for pathology detection, cleanings, restorative fillings, and complex extractions to alleviate acute oral pain and infection.
- Designed culturally sensitive workshops to **800+** international students, emphasizing the significance of oral health.

LEADERSHIP & ACTIVITIES

Undergraduate Bioinformatics Club | Publicity Chair

La Jolla, CA | Sept. 2024 - Present

- Brainstormed, filmed, and edited social media videos to triple the club's social media presence, totalling **50,000+** of views.
- Engaged in advanced workshops to open RNA sample data sets via Python and an intro to assembly algorithms.

Wesley United Methodist Church Mural | Project Lead

San Jose, CA | Aug. 2023 - Apr. 2024

- Spearheaded an **8-month** mural commission, managing end-to-end project lifecycles by coordinating **500+** volunteer hours and maintaining continuous communication with beneficiaries to integrate feedback into the final design.
- Engineered a comprehensive risk-mitigation strategy by researching environmental factors such as UV degradation and weather-related wear; collaborated with professional muralists to ensure the mural's long-term structural integrity.

SKILLS

Technical Skills: PCR & Gel Electrophoresis, Antibody Staining, Flow Cytometry (A3 & LSRII), FlowJo, Rodent Handling, Rodent Organ Dissection, Cell Culture, Python, Polars, Unix, R, Java, NumPy/Pandas, Sequence Alignment, HPC management

Interests: Video Creation, Basketball (Lakers), Volleyball, Thrifting, Sewing, Theatrical Arts, Musicals, Karaoke